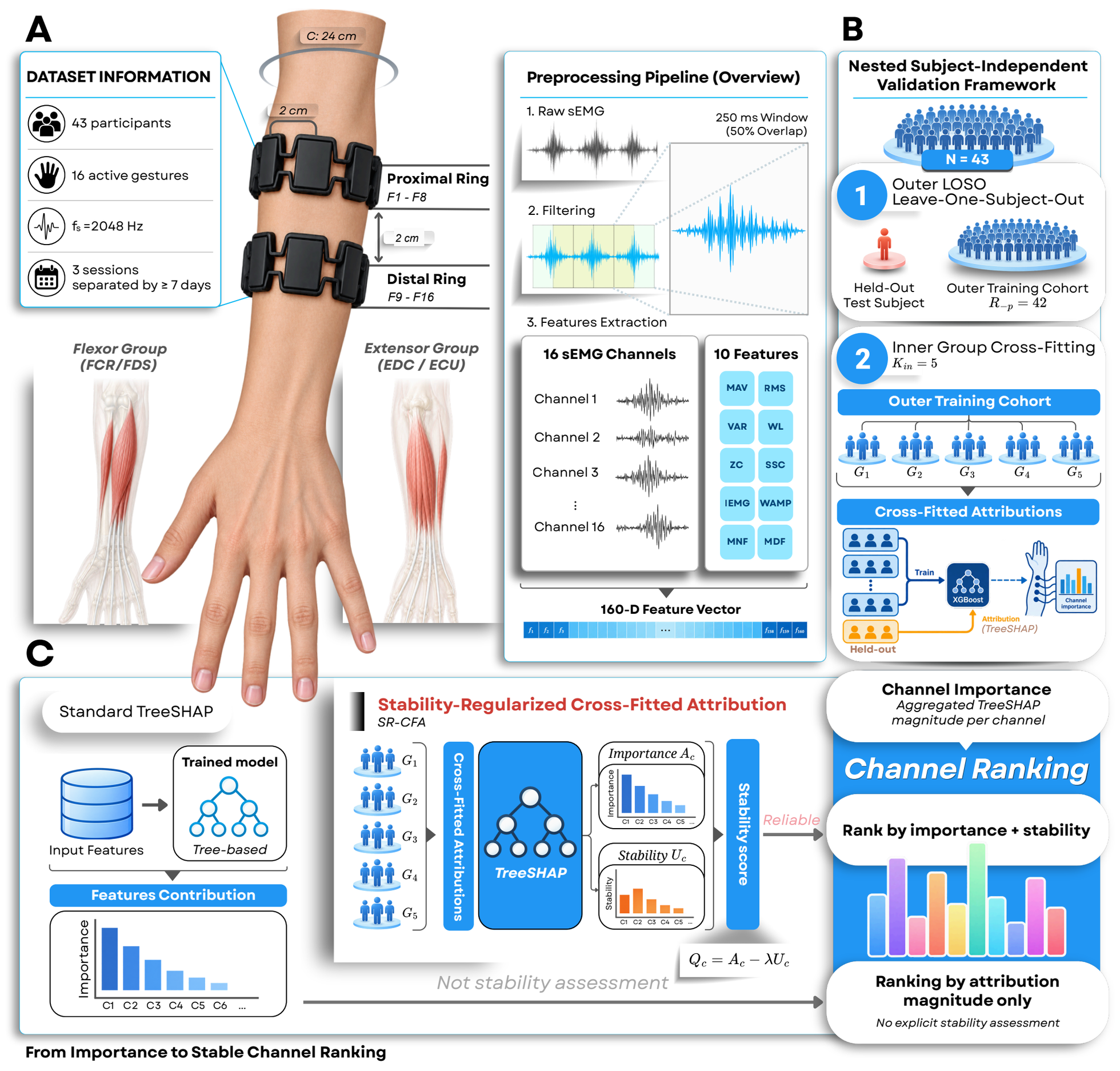

A
DATASET INFORMATION
43 participants
16 active gestures
f_s =2048 Hz
3 sessions separated by ≥ 7 days
C: 24 cm
2 cm
Proximal Ring
F1 - F8
2 cm
Distal Ring
F9 - F16
Flexor Group (FCR/FDS)
Extensor Group (EDC / ECU)
Preprocessing Pipeline (Overview)
1. Raw sEMG
250 ms Window (50% Overlap)
2. Filtering
3. Features Extraction
16 sEMG Channels
Channel 1
Channel 2
Channel 3
Channel 16
10 Features
MAV
RMS
VAR
WL
ZC
SSC
IEMG
WAMP
MNF
MDF
160-D Feature Vector
B
Nested Subject-Independent Validation Framework
N = 43
1 Outer LOSO Leave-One-Subject-Out
Held-Out Test Subject
Outer Training Cohort $R_{-p} = 42$
2 Inner Group Cross-Fitting $K_{in} = 5$
Outer Training Cohort
G_1 G_2 G_3 G_4 G_5
Cross-Fitted Attributions
Train
XGBoost
Channel importance
Attribution (TreeSHAP)
Held-out
C
Standard TreeSHAP
Input Features
Trained model
Tree-based
Features Contribution
Importance
C1 C2 C3 C4 C5 C6 ...
Stability-Regularized Cross-Fitted Attribution
SR-CFA
G_1 G_2 G_3 G_4 G_5
Cross-Fitted Attributions
TreeSHAP
Importance A_c
Stability U_c
Stability score
Reliable
$Q_c = A_c - \lambda U_c$
Not stability assessment
Channel Importance
Aggregated TreeSHAP magnitude per channel
Channel Ranking
Rank by importance + stability
Ranking by attribution magnitude only
No explicit stability assessment
From Importance to Stable Channel Ranking